JM0211 Process Mining

Conformance Checking for Process Improvement

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Motivation and Scientific Background

- **Use case.** A Dutch bank (BPI Challenge 2017) denied 3,093 loan applications, in under a day or after about 260 days.
- **Question.** Which deviations from the normative process go with slow denials? Allow or forbid them?
- **Why not flag all deviations?** Some are workarounds linked to better performance (Dees et al., 2017).
- **Course fit.** Resource and temporal conformance check text constraints; we check **control flow against a model**.

3,093 denied applications

54 visible activities

14.43 days, median throughput

fast 1,501 | slow 1,592

cutoff 14 days

Explanation of Conceptual Approach

Input **XES log**: case id, activity, timestamp per event.
PNML Petri net: normative model.

Output **Rules** on move counts, plus the **traces** satisfying each rule (CSV).

1 KPI Throughput time per trace, in days. ›

2 Align, encode Optimal alignment; count moves. ›

3 Classify Slow above cutoff; decision tree. ›

4 Rules Each leaf is one rule.

Step 2: trace A, A, C vs model A, B, C

	1	2	3	4
Trace	A	A	>>	C
Model	A	>>	B	C
Move	sync	**log move A**	**model move B**	sync

`log_move_A = 1, model_move_B = 1 → 54 x 2 = 108 features`

Parameters

Cutoff	14 days; slow above
Hold-out	25%, seed 0
Grid	48: depth, leaf, pruning
Tuning	5-fold CV, F1 of slow

Evaluation Design and Results

1. Correctness: hand-built gold standard

2 toy logs (5 and 6 traces), move counts and a planted rule derived by hand first.

5/5, 6/6 traces exact | **1.0** alignment P, R | **1.0** rule P, R

2. Usefulness: real log, 25% hold-out

Precision (P), recall (R), F1 of the slow class vs an **all-slow baseline**, seeds 0 to 3.

0.712 F1 seed 0 | **0.688** mean | **0.680** baseline

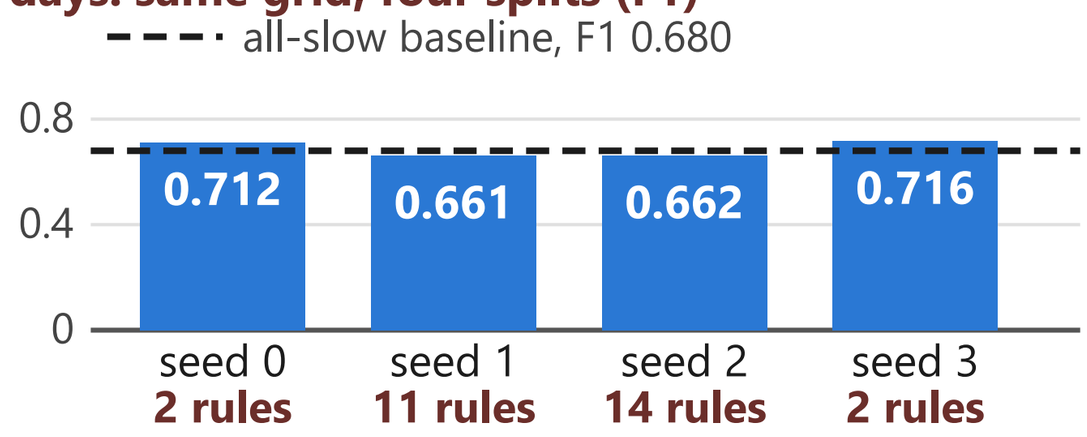
Seed 0 tree: two rules, one split

log move of `W-Call-after-offers-suspend` ?

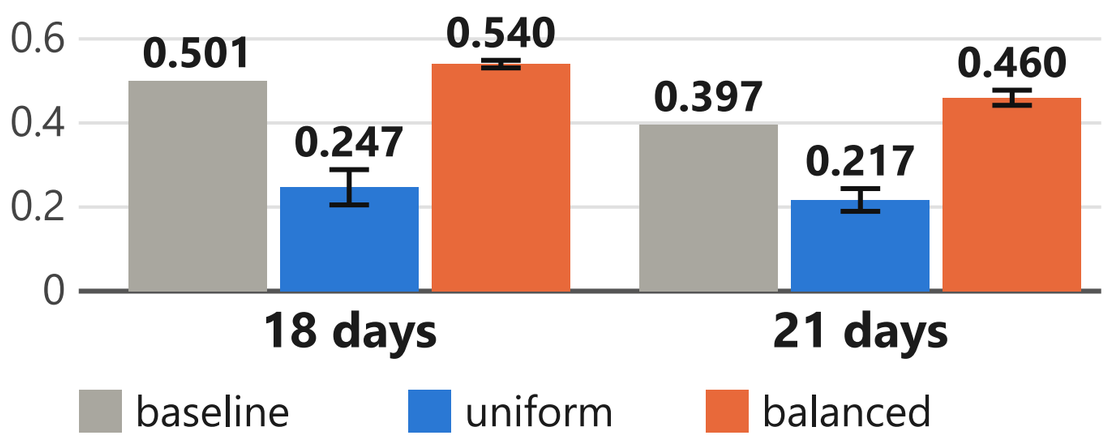
- **none** → **389** traces, 16% slow → **predict fast**
- **at least one** → **2,704** traces, 57% slow → **predict slow**

Base rate of slow: 51.5%. P 0.563, R 0.970, F1 0.712

14 days: same grid, four splits (F1)



18 and 21 days: class weights (F1)



Correct on the gold standard, but the deviation-to-delay link is weak and unstable.

Limitations and Future Work

Limitations

- Two-rule tree is unstable across splits.
- Root split hits 87% of traces: net may be too strict.
- One cutoff, one dataset, control flow only.

Future work

- Regression tree on continuous throughput time.
- Sample several optimal alignments per trace.
- Data-aware checking (guards such as loan amount).

References: Dees, de Leoni, Mannhardt (2017). van der Aalst (2016) Process Mining. Berti et al. (2019) PM4Py. van Dongen (2017).